To-do

Six-week paper plan

Paper aim

To test whether culturally patterned social cognition in Chinese- and Western-developed multimodal AI models reflects:

1. model-development ecosystem;
2. language-dependent elicitation;
3. experimental condition;
4. differences across model generations.

Week 1: Freeze the study design

Kangning

- Refine the theoretical framework:
    - cultural imprinting;
    - contextual adaptation;
    - cross-cultural convergence.
- Define the psychological construct being measured.
- Confirm whether the questionnaire should be described as:
    - theory of mind;
    - visual mental-state inference;
    - social cognition.
- Review the psychological validity and limitations of the questionnaire.
- Draft the research questions and hypotheses.
- Review the Chinese translation of instructions and response options.
- Identify terms that are psychologically or culturally difficult to translate.

Ming

- Compile the proposed model list.
- Confirm which models:
    - accept image inputs;
    - are accessible through API or interface;
    - have historical versions available;
    - support Chinese and English prompts.
- Document model names, providers, release dates and access methods.
- Freeze the image preprocessing and encoding workflow.
- Prepare the bilingual prompting templates.
- Confirm that each request begins in a fresh context.
- Test response extraction under both Chinese and English outputs.

Jie

- Finalise the factorial design:
    - ecosystem;
    - provider;
    - model;
    - generation;
    - language;
    - condition;
    - item;
    - repetition.
- Define the primary and secondary outcomes.
- Write the statistical analysis plan.
- Define the minimum repetitions per cell.
- Create the master data dictionary.
- Prepare the project tracker and manuscript outline.
- Confirm the target journal and article type.

Week 1 deliverables

- Final model list.
- Final hypotheses.
- Final bilingual prompts.
- Final analysis plan.
- Final data structure.
- Pilot-ready experimental pipeline.

Week 2: Pilot and quality assurance

Kangning

- Review a sample of English and Chinese model responses.
- Check whether translated mental-state words retain the intended distinctions.
- Identify ambiguous items or response options.
- Develop rules for interpreting psychologically ambiguous responses.
- Draft the theoretical background section of the Introduction.

Ming

- Run a pilot across:
    - at least two Chinese-developed models;
    - at least two Western-developed models;
    - both languages;
    - both conditions.
- Test randomisation of response-option order.
- Check API stability, delays, retries and logging.
- Validate that the original response, extracted response and errors are all retained.
- Identify models requiring provider-specific technical handling.
- Estimate time and cost for full data collection.

Jie

- Review pilot completeness and response distributions.
- Check for:
    - excessive invalid responses;
    - extraction failures;
    - option-position effects;
    - near-deterministic items;
    - ceiling or floor effects.
- Fit a preliminary mixed-effects model.
- Decide whether five or ten repetitions per cell are needed.
- Revise the analysis plan based on the pilot without changing the core hypotheses.

Week 2 deliverables

- Pilot dataset.
- Pilot quality-control report.
- Final extraction rules.
- Final repetition number.
- Locked experimental protocol.

Week 3: Main data collection

Kangning

- Continue drafting the Introduction.
- Write the theoretical accounts and hypotheses.
- Prepare the psychological interpretation framework for the Discussion.
- Develop a terminology guide to ensure consistent use of:
    - cultural imprinting;
    - ecosystem effects;
    - language effects;
    - visual mental-state inference.
- Review any unexpected response patterns identified during collection.

Ming

- Run the full experiment across all eligible models.
- Monitor progress by:
    - model;
    - language;
    - condition;
    - item;
    - repetition.
- Re-run failed or incomplete requests according to the prespecified retry rules.
- Record exact timestamps and model identifiers.
- Maintain a log of:

    - model changes;
    - API outages;
    - refusals;
    - unsupported functions;
    - altered model access.
- Export a complete raw dataset and execution log.

Jie

- Monitor completeness without examining primary inferential results prematurely.
- Produce daily checks for:
    - missing cells;
    - duplicate records;
    - extraction errors;
    - model imbalance.
- Ensure no model family dominates the ecosystem comparison.
- Prepare the analysis scripts using the pilot data.
- Draft the Methods sections on:
    - study design;
    - outcomes;
    - statistical analysis.

Week 3 deliverables

- Complete raw experimental dataset.
- Data-collection log.
- Model metadata table.
- Draft Introduction.
- Draft statistical Methods.

Week 4: Data cleaning and primary analysis

Kangning

- Review unresolved and ambiguous responses.
- Help classify errors as:
    - perceptual or conceptual error;
    - inappropriate mental-state inference;
    - linguistic ambiguity;
    - response-format failure.
- Review whether item-level findings are psychologically interpretable.
- Draft the conceptual limitations section.

Ming

- Finalise the response-processing pipeline.
- Produce the analysis-ready dataset.

- Verify all data transformations against raw responses.
- Summarise:
    - API errors;
    - invalid responses;
    - refusals;
    - extraction fallbacks.
- Write the technical Methods sections:
    - model access;
    - prompting;
    - image handling;
    - response extraction;
    - error handling.

Jie

- Run descriptive analyses.
- Fit the primary item-level mixed-effects model.
- Estimate:
    - ecosystem effect;
    - language effect;
    - ecosystem-by-language interaction;
    - condition effect;
    - generation effects.
- Run provider-level analyses.
- Analyse response consistency and invalid-response rates.
- Produce estimated marginal probabilities and confidence intervals.
- Check whether non-linear trajectory claims are statistically supportable.
- Begin drafting Results.

Week 4 deliverables

- Locked analysis dataset.
- Primary statistical results.
- Provider-level results.
- Initial figures and tables.
- Complete Methods draft.
- Initial Results draft.

Week 5: Sensitivity analyses and manuscript integration

Kangning

- Interpret the findings against the competing theoretical accounts.
- Determine whether results support:
    - cultural imprinting;
    - contextual adaptation;
    - convergence;

    - a hybrid account.
- Draft the Discussion.
- Ensure claims do not overstate:
    - culture;
    - nationality;
    - cognition;
    - theory of mind.
- Review the manuscript for psychological accuracy.

Ming

- Run technical sensitivity checks:
    - exact extraction only;
    - fallback extraction included;
    - invalid responses treated as incorrect;
    - invalid responses excluded;
    - option-order effects;
    - models with incomplete API comparability removed.
- Prepare supplementary materials:
    - prompts;
    - response schema;
    - model metadata;
    - error logs;
    - code documentation.
- Prepare the reproducibility repository.

Jie

- Run statistical sensitivity analyses.
- Compare:
    - pooled ecosystem results;
    - provider-stratified results;
    - model-family results;
    - generation-specific contrasts.
- Finalise all figures and tables.
- Integrate Introduction, Methods, Results and Discussion.
- Draft the abstract and conclusion.
- Ensure the manuscript matches Applied Artificial Intelligence's scope.

Week 5 deliverables

- Full manuscript draft.
- Final primary figures.
- Supplementary analyses.
- Reproducibility materials.
- Draft abstract.

Week 6: Internal review and submission

Kangning

- Conduct a full theory and psychology review.
- Tighten the Introduction and Discussion.
- Verify that cultural interpretations are appropriately cautious.
- Check all questionnaire and psychological references.
- Approve the final terminology and theoretical claims.

Ming

- Conduct a full technical reproducibility review.
- Verify:
    - prompts;
    - code;
    - model identifiers;
    - API settings;
    - timestamps;
    - image handling.
- Confirm that another researcher could reproduce the experiment.
- Finalise the repository and supplementary technical appendix.

Jie

- Conduct the final statistical and editorial review.
- Check all reported numbers against outputs.
- Finalise:
    - title;
    - abstract;
    - highlights;
    - keywords;
    - cover letter;
    - author contributions;
    - data and code availability statements.
- Format the manuscript for Applied Artificial Intelligence.
- Submit the paper and post the preprint.

Week 6 deliverables

- Submission-ready manuscript.
- Cover letter.
- Supplementary files.
- Public or review-ready code repository.
- Preprint.
- Submission to Applied Artificial Intelligence.

Lead responsibility by manuscript section

Section	Lead	Support
Theoretical framework	Kangning	Jie
Psychological background	Kangning	Jie
Experimental implementation	Ming	Jie
Model and API documentation	Ming	Jie
Statistical design	Jie	Kangning
Data analysis	Jie	Ming
Introduction	Kangning	Jie
Methods	Ming and Jie	Kangning
Results	Jie	Ming
Discussion	Kangning and Jie	Ming
Supplementary technical material	Ming	Jie
Final manuscript integration	Jie	All

Weekly meeting structure

Hold one fixed 30-minute meeting each week.

Standing agenda

1. Tasks completed.
2. Current blockers.
3. Decisions required.
4. Data or manuscript quality issues.
5. Tasks for the following week.

During the data-collection period, Ming should also provide a brief written progress update showing:

- completed model–language–condition cells;
- failed cells;
- models with access problems;
- expected remaining runs.

Scope rules

To keep the submission fast:

- Do not add human recruitment.
- Do not create a completely new visual benchmark.
- Do not continually add new models after the model list is frozen.
- Do not fit complex non-linear models unless model-generation coverage supports them.
- Do not make “Chinese-like” or “Western-like” claims without human normative data.
- Prioritise a complete, reproducible design over maximum model coverage.

Plan

Paper plan

Working title

Cultural Imprinting or Contextual Adaptation? A Cross-Linguistic Evaluation of Social Cognition in Chinese- and Western-Developed Multimodal AI Models

Central aim

To determine whether culturally patterned differences in multimodal artificial social cognition reflect:

1. persistent model-development ecosystem effects;
2. language-dependent elicitation;
3. differences across model generations;
4. interactions between these factors.

Core research question

Do Chinese- and Western-developed multimodal AI models show different visual mental-state inference patterns, and do these differences persist or change when the same models are tested in Chinese and English?

Minimum viable study

Use the existing photographic questionnaire and current data-collection pipeline.

Include:

- Chinese-developed multimodal model families;
- Western-developed multimodal model families;
- at least two model generations per major provider where accessible;
- Chinese and English administration;
- both existing experimental conditions;
- repeated independent responses;
- randomised response-option order;
- complete retention of invalid responses and errors.

The minimum factorial structure is:

Development ecosystem × provider × prompt language × condition × model generation × item × repetition

Model selection

Aim for at least:

Chinese-developed models

Three model families, for example:

- Qwen
- ERNIE
- GLM or another accessible Chinese multimodal family

Western-developed models

Three model families, for example:

- GPT
- Claude
- Gemini

Only include models that can receive the same photographic input directly.

For each model, record:

- exact model identifier;
- provider;
- release date;
- model family;
- API or platform;
- data-collection date;
- temperature;
- top-p;
- maximum output tokens;
- image resolution;
- system prompt;
- user prompt;
- structured-output settings.

Experimental protocol

Primary administration

Every model receives every item:

- in English;
- in Chinese;
- under condition 1;
- under condition 2;
- across repeated independent runs.

Each request should begin in a fresh conversation or context.

Repetitions

Target:

- 10 repetitions per model–item–language–condition combination, if computationally affordable;
- minimum 5 repetitions where cost or API access is restrictive.

Randomisation

Randomise:

- item order;
- response-option order;
- language block order;
- condition order.

Keep the photograph itself unchanged across languages.

Translation

The Chinese version should be developed using:

1. forward translation;
2. independent back-translation;
3. bilingual reconciliation;
4. checking that the mental-state words retain comparable meaning;
5. pilot testing on a small number of items and models.

Document any response options that do not have a clean one-to-one translation.

Primary outcomes

Accuracy

Whether the extracted answer matches the designated correct response.

Response consistency

How often the same answer is selected across repetitions.

Useful measures:

- modal-response proportion;
- probability of selecting the correct option;
- within-item response entropy.

Invalid-response rate

Whether the response:

- cannot be mapped to an option;

- contains several options;
- refuses the task;
- produces an API or processing failure.

Primary hypotheses

H1: Ecosystem difference

Chinese- and Western-developed model families differ in visual mental-state inference after accounting for language, condition, provider and generation.

H2: Language effect

The same models respond differently when the questionnaire is administered in Chinese versus English.

H3: Ecosystem-by-language interaction

The effect of prompt language differs between Chinese- and Western-developed model families.

H4: Generational change

The cultural gap changes across successive model generations.

H5: Accuracy–consistency distinction

Newer models may become more accurate without becoming proportionally more stable.

Statistical analysis

Primary model

Fit an item-level mixed-effects logistic regression for correctness.

Conceptually:

Correctness ~ ecosystem × language + condition + generation + selected interactions + random effects

Random effects should include:

- item;
- model family;
- model version;
- repeated model–item observations where appropriate.

Key interaction tests

Prioritise:

- ecosystem × language;
- ecosystem × generation;
- language × generation;
- ecosystem × language × generation, if supported by the number of models.

Do not begin with an over-complex full interaction model.

Secondary analyses

Analyse:

- invalid-response probability;
- response consistency;
- response entropy;
- option-position effects;
- provider-specific results;
- sensitivity to excluding unresolved responses;
- sensitivity to deterministic versus stochastic settings.

Non-linearity

Only claim non-linear development where there are enough generations to support it.

Where model histories are short, report:

- generation-specific contrasts;
- step changes;
- plateaus;
- regressions.

Avoid fitting smooth curves to two or three model versions.

Planned figures

Figure 1

Study design and conceptual framework:

- development ecosystem;
- prompt language;
- model generation;
- experimental condition;
- observed social-cognitive performance.

Figure 2

Provider-level accuracy across model generations.

Figure 3

Chinese versus English performance for each model family.

Figure 4

Ecosystem-by-language interaction using estimated marginal probabilities.

Figure 5

Accuracy and response consistency across generations.

Figure 6

Invalid-response and error rates by provider and language.

Planned tables

Table 1

Model characteristics:

- provider;
- ecosystem;
- model family;
- version;
- release date;
- API date;
- number of items;
- number of repetitions.

Table 2

Overall performance by model, language and condition.

Table 3

Mixed-model estimates and key interaction contrasts.

Supplementary tables

- item-level accuracy;
- unresolved responses;
- option-order sensitivity;
- full prompts;
- translation mapping;
- sensitivity analyses.

Manuscript structure

Introduction

1. Multimodal AI is increasingly used for socially sensitive tasks.
2. Artificial social cognition is often evaluated as though it were culturally neutral.
3. Model behaviour may reflect training ecosystem, language of elicitation and model generation.
4. Existing studies have examined these influences separately.
5. This study jointly evaluates development ecosystem, language and generation using a fully crossed design.
6. Present the hypotheses.

Methods

- Model selection
- Experimental materials
- Chinese translation
- Experimental conditions
- Prompting protocol
- Repetitions and randomisation
- Response extraction
- Outcomes
- Statistical analysis
- Reproducibility

Results

1. Data completeness
2. Overall model performance
3. Provider-level differences
4. Ecosystem differences
5. Language effects
6. Ecosystem-by-language interaction
7. Generational change
8. Consistency and invalid responses
9. Sensitivity analyses

Discussion

1. Is artificial social cognition culturally neutral?
2. Evidence for ecosystem imprinting
3. Evidence for language-dependent adaptation
4. Whether models converge across generations
5. Implications for culturally portable AI
6. Limitations
7. Future work using culturally matched visual stimuli

Interpretation boundaries

Use:

- visual mental-state inference;
- social-cognitive task performance;
- culturally patterned model behaviour;
- model-development ecosystem;
- ecosystem-associated differences.

Avoid:

- Chinese models think like Chinese people;
- Western models possess Western cognition;
- model nationality causes the observed behaviour;
- questionnaire accuracy proves genuine theory of mind.

Minimum publishable claim

Chinese- and Western-developed multimodal model families show different language-conditioned trajectories in visual mental-state inference.

Stronger claim

Artificial social cognition reflects both persistent development-ecosystem differences and flexible language-dependent adaptation, with the balance between these influences changing across model generations.

Submission strategy

Target journal

Applied Artificial Intelligence

Article type

Research Article

Dissemination

Post a preprint at submission and archive:

- prompts;
- model metadata;
- analysis code;
- response-processing code;
- de-identified or licence-compliant response data.

Draft

Journal: Applied Artificial Intelligence

Title: Cultural Imprinting or Contextual Adaptation? A Cross-Linguistic Evaluation of Social Cognition in Chinese- and Western-Developed Multimodal AI Models

[Author names]

[Affiliations]

Corresponding author: [Name and email]

Abstract

Multimodal artificial intelligence models are increasingly expected to interpret socially meaningful visual information across languages and cultural settings. However, current evaluations commonly treat artificial social cognition as culturally neutral or compare models using only one language. It therefore remains unclear whether culturally patterned model behaviour reflects persistent influences associated with the model-development ecosystem, flexible adaptation to the language of elicitation, or broader changes across model generations.

We evaluated [N] multimodal models from [N] Chinese-developed and [N] Western-developed model families using a visual mental-state inference task. A fully crossed design was used in which every eligible model responded to every item under two experimental conditions and in both Chinese and English. Each model–item–language–condition combination was repeated [N] times, producing [N] attempted responses. Responses were retained in full and processed using a prespecified extraction procedure. The primary outcome was item-level accuracy, with response consistency and unresolved-response frequency examined as secondary outcomes.

Item-level mixed-effects models were used to estimate the effects of model-development ecosystem, prompt language, experimental condition and model generation while accounting for variation among items, providers, model families and repeated responses. [Insert one sentence reporting the principal ecosystem result.] [Insert one sentence reporting the ecosystem-by-language result.] [Insert one sentence reporting the generational finding.] [Insert one sentence reporting consistency or invalid-response findings.]

The findings indicate that artificial social cognition is [culturally situated/context-sensitive/partially convergent] rather than uniformly culturally neutral. Observed performance reflected [persistent ecosystem-associated differences, language-dependent elicitation, and/or their interaction], with these influences [changing/remaining stable] across model generations. These results demonstrate the importance of evaluating socially oriented multimodal AI across both development ecosystems and languages before deployment in culturally diverse settings.

Keywords: multimodal artificial intelligence; social cognition; theory of mind; cultural alignment; vision-language models; Chinese language; cross-cultural AI; model evaluation

1. Introduction

Multimodal artificial intelligence models are increasingly used in settings requiring the interpretation of human expressions, intentions and socially meaningful visual information. As these systems are

deployed across countries and languages, their apparent capacity to infer mental states may influence human–AI interaction, education, healthcare, customer support and other socially sensitive applications. Reliable performance in one linguistic or cultural context does not, however, guarantee that the same capability will generalise across contexts.

Evaluations of artificial social cognition have expanded rapidly. Language models have demonstrated substantial performance on tasks involving false beliefs, indirect requests, social inferences and other constructs commonly associated with theory of mind. More recent multimodal evaluations have extended this work to photographs, videos and narrative-rich social scenarios. These studies suggest that contemporary models can extract socially relevant information from visual input, although performance varies substantially across models, task types and modes of presentation.

One widely used assessment is the Reading the Mind in the Eyes Test and related visual mental-state inference tasks. These instruments require the respondent to select the mental-state term that best describes a photograph of the eye region. Recent evaluations indicate that some advanced multimodal models can perform strongly on such tasks, while earlier or alternative models may perform close to chance. Performance has also been examined across stimulus sets representing different racial or ethnic groups. These findings demonstrate rapid changes in model capability, but they do not establish whether model performance is culturally neutral or whether different model-development ecosystems produce distinct patterns of social inference.

Culture may enter model behaviour through several pathways. Training corpora contain culturally structured language, social norms and representations of interpersonal behaviour. Post-training, alignment and safety procedures may further influence how models interpret socially ambiguous information. These processes occur within particular commercial, linguistic and regulatory environments. Models developed primarily within China and those developed primarily within the United States or Europe, may therefore be exposed to different combinations of training data, annotation practices, alignment objectives and deployment expectations.

The language used to interact with a model provides a second potential source of cultural variation. Recent evidence shows that the same generative model can exhibit more interdependent and holistic response tendencies when prompted in Chinese than when prompted in English. Similar tendencies have been observed in both Western-developed and Chinese-developed language models. These results suggest that model behaviour is not determined entirely by provider or training origin. Instead, language may activate different learned representations and response tendencies within the same model.

Multimodal cultural evaluation introduces a third layer of complexity. Contemporary vision-language models show unequal knowledge and performance across cultural regions, and their responses can be sensitive to linguistic reformulation and visual presentation. Models may therefore demonstrate culturally patterned performance not only because of the language used in the prompt, but also because of how visual and linguistic information are integrated. This is particularly important for social cognition, where mental-state judgements may depend on subtle visual cues whose interpretation is culturally situated.

Despite these developments, three limitations remain. First, studies of model culture have generally focused on textual values, reasoning styles or cultural knowledge rather than visually grounded mental-state inference. Second, multimodal theory-of-mind studies have primarily compared individual models or stimulus groups without systematically separating model-development ecosystem from elicitation language. Third, models are commonly treated

cross-sectionally, even though the effects of language and the development ecosystem may change substantially across successive model generations.

The present study addresses these limitations through a fully crossed evaluation of Chinese- and Western-developed multimodal model families. Every eligible model was tested using the same photographic items under two experimental conditions and in both Chinese and English. Multiple generations were included where accessible, and each model–item–condition combination was repeated to distinguish average accuracy from response stability.

We considered three theoretical accounts.

The **cultural-imprinting account** predicts persistent differences between Chinese- and Western-developed model families after prompt language and experimental condition are held constant. Under this account, model behaviour retains systematic influences associated with the ecosystem in which the model was developed.

The **contextual-adaptation account** predicts that the apparent cultural orientation of a model changes according to the language in which the task is administered. Under this account, cultural behaviour is flexibly elicited rather than fixed at the model-family level.

The **convergence account** predicts that ecosystem- and language-associated differences decrease across successive model generations as multimodal social-inference capabilities become more generalisable.

These accounts are not mutually exclusive. A model may retain persistent ecosystem-associated tendencies while simultaneously adapting its responses to the language of administration. The relative contribution of these processes may also vary across model generations.

We tested the following hypotheses:

H1: Development-ecosystem differences. Chinese- and Western-developed model families differ in visual mental-state inference after accounting for prompt language, experimental condition, provider and model generation.

H2: Language-dependent elicitation. The same models produce different response patterns when tested in Chinese rather than English.

H3: Ecosystem-by-language interaction. The effect of prompt language differs between Chinese- and Western-developed model families.

H4: Generational change. Ecosystem- and language-associated differences change across successive model generations.

H5: Accuracy and consistency are distinct. Changes in mean accuracy are not necessarily accompanied by equivalent changes in response consistency.

Rather than interpreting questionnaire performance as evidence that AI systems possess human-like mental states, we examine demonstrated performance on a structured visual mental-state inference task. This distinction permits a controlled comparison of model behaviour without making claims about consciousness or psychological mechanisms that cannot be established from behavioural outputs alone.

2. Materials and Methods

2.1 Study design

The study used a fully crossed repeated-measures design. Every eligible multimodal model responded to every item under both experimental conditions and in both English and Chinese. Each model–item–language–condition combination was administered [N] times using independent requests.

The principal experimental factors were:

- model-development ecosystem: Chinese-developed or Western-developed;
- provider;
- model family;
- model version or generation;
- prompt language: Chinese or English;
- experimental condition: [Condition 1] or [Condition 2];
- questionnaire item;
- repetition.

The condition under which each response was elicited was recorded for every observation, enabling direct within-model and within-item comparisons.

2.2 Model eligibility and classification

Models were eligible when they:

1. could directly process the photographic input;
2. could receive substantively equivalent prompts in Chinese and English;
3. were publicly accessible through a documented application programming interface or reproducible platform during the study period;
4. could return a textual response that could be mapped to the permitted response options.

The final analysis included [N] models from [N] providers. Chinese-developed models comprised [list model families and versions], and Western-developed models comprised [list model families and versions].

Models were classified by the principal location and organisational ecosystem of their developer rather than by the presumed culture of the model itself. The terms “Chinese-developed” and “Western-developed” are therefore descriptive classifications of development ecosystem and should not be interpreted as direct measures of cultural identity.

For each model, we recorded:

- provider;
- exact model identifier;
- model family;
- publicly reported release date;
- date and time of data collection;
- application programming interface or platform;

- model-development ecosystem;
- image-input format and resolution;
- system and user prompts;
- temperature and other sampling parameters;
- maximum output length;
- structured-output settings;
- request and retry rules.

A full description of the included models is provided in Table 1.

Table 1. Characteristics of included multimodal models

Ecosystem	Provider	Model family	Exact model identifier	Release date	Collection date	Repetitions
Chinese-developed	[Provider]	[Family]	[Identifier]	[Date]	[Date]	[N]
Western-developed	[Provider]	[Family]	[Identifier]	[Date]	[Date]	[N]

2.3 Visual mental-state inference task

The study used [full name of questionnaire or task], which comprises [N] photographic items. For each item, the model was shown [describe photograph, such as a cropped image of the eye region] and asked to select the term that best described the depicted mental state from [N] response options.

Each item included:

- a unique item identifier;
- a photographic stimulus;
- a prespecified set of response options;
- a designated correct answer.

The task was used as an assessment of **visual mental-state inference**. Although this capability is relevant to theory of mind and social cognition, task accuracy should not be interpreted as a comprehensive measure of human-like theory of mind.

[Add a paragraph describing permission, licensing or authorised access to the questionnaire images. State whether images can be redistributed.]

2.4 Experimental conditions

Each item was administered under two conditions.

Condition 1: [Condition name].
[Describe exactly what information was shown and what instruction was used.]

Condition 2: [Condition name].
[Describe exactly what information was shown and how it differed from Condition 1.]

The underlying image, response options and designated correct answer were held constant across conditions unless the condition definition explicitly required otherwise. The condition was recorded for every response.

The difference between conditions was intended to test [state psychological or technical rationale—for example, whether explicit response scaffolding improves visual mental-state inference].

2.5 Chinese and English administration

All task instructions and response options were administered in both Chinese and English. The English version was based on [original questionnaire/instrument wording]. The Chinese version was developed through [forward translation, independent review and/or back-translation].

Translation focused on conceptual rather than strictly literal equivalence. Particular attention was given to mental-state terms that lacked a simple one-to-one correspondence across languages. Disagreements were reviewed by [number] bilingual researchers with expertise in [psychology/language/cross-cultural research]. The final translation mapping is provided in Supplementary Table S1.

Photographic stimuli were identical across prompt languages. Thus, the language comparison reflects a change in the linguistic elicitation context rather than a change in the visual input.

2.6 Prompting procedure

A standardised prompt template was used across all providers wherever technically possible. Each request contained:

1. the photographic stimulus;
2. the task instruction;
3. the response options;
4. an instruction to return one selected word or option.

The English prompt was:

[Insert final English prompt.]

The Chinese prompt was:

[Insert final Chinese prompt.]

Each request began in a fresh conversational context to prevent previous items or responses from influencing subsequent outputs. No corrective feedback was provided.

Provider-specific modifications were limited to technical syntax required for image input or structured output. Any such modifications are documented in Supplementary Table S2.

Response-option order was [randomised/fixed]. Where randomised, the order presented to the model was retained for each observation, permitting assessment of option-position effects.

A brief delay of [N] seconds was imposed between successive requests to support stable collection and reduce rate-limiting errors. Requests that failed for technical reasons were retried according to the prespecified rule of [describe retry rule]. Original failures and retry attempts were retained in the execution log.

2.7 Image processing

To improve computational efficiency and maintain consistency, each photographic stimulus was encoded once in the format required by the relevant provider and reused across eligible models and repetitions rather than reprocessed for every request.

Images were presented at [resolution and format]. No image enhancement, alteration or cropping was performed beyond [describe any required preprocessing]. Where provider-specific encoding was necessary, visual content and dimensions were held as consistent as technically possible.

2.8 Repeated administrations

Each model completed [N] independent repetitions of every item in each language and experimental condition. Repeated administrations were used to estimate both expected accuracy and response stability.

Requests were sampled using [temperature] and [top-p or equivalent]. The primary protocol used [low/deterministic/moderate] sampling to reduce unnecessary response variation while retaining [brief rationale]. [If applicable: A secondary stochastic condition was used as a sensitivity analysis.]

2.9 Response extraction and validation

The complete original output was retained for every request. Each output was then processed to identify the selected response option.

Outputs were first normalised into a consistent textual form. The processing routine then attempted to extract the selected word from the expected structured response and match it against the permitted options. Where structured extraction was unsuccessful, a fallback procedure searched the response text for occurrences of valid option words.

A response was classified as unresolved when:

- no permitted option could be identified;
- multiple conflicting options were identified and could not be disambiguated;
- the model refused to answer;
- a technical or processing error prevented classification.

Responses that could not be resolved, together with collection and processing errors, were explicitly flagged rather than discarded. This preserved a complete record of all attempted model–item evaluations.

For every observation, the dataset retained:

- item identifier;
- image identifier;
- response options and their presented order;
- correct answer;

- experimental condition;
- prompt language;
- provider and model identifier;
- repetition number;
- extracted response;
- accuracy indicator;
- original model output;
- extraction method;
- unresolved-response indicator;
- request or processing error;
- timestamp.

A random sample of [N or percentage] responses was manually reviewed to evaluate extraction accuracy. Agreement between the automated and manually reviewed classifications was [insert agreement statistic].

2.10 Outcomes

2.10.1 Primary outcome

The primary outcome was item-level correctness, defined as whether the extracted response matched the designated correct answer.

2.10.2 Response consistency

Consistency was evaluated across repeated administrations of the same model–item–language–condition combination using:

- modal-response proportion;
- proportion of repetitions returning the correct response;
- within-cell response entropy.

Lower entropy indicated greater consistency, while higher entropy indicated greater variability across repeated requests.

2.10.3 Unresolved responses

A secondary binary outcome indicated whether a response could be mapped unambiguously to one permitted option. Unresolved responses were examined separately because failure to follow the output format is conceptually distinct from selection of an incorrect mental-state option.

2.11 Statistical analysis

Descriptive summaries were produced for every model, language and experimental condition. These included accuracy, consistency, unresolved-response frequency and corresponding confidence intervals.

The primary analysis used item-level mixed-effects logistic regression, with correctness as the binary outcome. The prespecified fixed effects included:

- model-development ecosystem;
- prompt language;

- experimental condition;
- model generation or release order;
- ecosystem-by-language interaction;
- ecosystem-by-generation interaction;
- language-by-generation interaction.

The ecosystem-by-language interaction was the primary test distinguishing persistent development-ecosystem differences from language-dependent adaptation.

Random effects were used to account for clustering by:

- questionnaire item;
- provider or model family;
- exact model version;
- repeated observations within model–item combinations.

The initial model was specified as:

$$
[
$$

$$
\operatorname{logit}\{P(Y=1)\}
$$

$$
\beta_0+
 \beta_1E+
 \beta_2L+
 \beta_3C+
 \beta_4G+
 \beta_5(E\times L)+
 \beta_6(E\times G)+
 \beta_7(L\times G)
 +
 u_{\text{item}}
 +
 u_{\text{family}}
 +
 u_{\text{model}},
 ]
$$

where (E) denotes development ecosystem, (L) denotes prompt language, (C) denotes experimental condition and (G) denotes model generation.

Generation was represented using [ordered within-family generation/release date/generation-specific categorical effects]. Because model labels are not comparable across providers, generation was not interpreted as a common numerical scale unless supported by the available data.

A three-way ecosystem-by-language-by-generation interaction was fitted only where model coverage was sufficient to estimate it reliably.

Results are reported as adjusted predicted probabilities, probability differences or odds ratios with 95% confidence intervals. Provider-level estimates accompany pooled ecosystem estimates to ensure that an apparent ecosystem difference is not driven by one provider.

Separate models were fitted for response consistency and unresolved-response frequency. [State model type for entropy or consistency.]

Sensitivity analyses:

1. classified unresolved responses as incorrect;
2. excluded unresolved responses;
3. restricted extraction to responses resolved through the primary structured procedure;
4. excluded models lacking directly comparable sampling settings;
5. adjusted for response-option position;
6. repeated analyses separately by provider;
7. compared [deterministic and stochastic settings, if applicable].

Evidence for non-linearity was assessed by comparing [linear, categorical, segmented or spline] representations of generation. Non-linear trajectories were reported only when the number and spacing of model versions provided adequate support.

All analyses were conducted using [software and version]. Analysis code was developed before final inferential results were examined and is available at [repository].

2.12 Ethical considerations

The study evaluated outputs from publicly or institutionally accessible AI models and did not involve human participants or identifiable personal data. [Insert institutional ethics determination if one was sought.]

The study nevertheless raises ethical issues concerning cultural categorisation and interpretation. Development ecosystem was treated as a property of the model provider and training context, not as evidence that a model represents all people from a country or culture.

2.13 Reproducibility

The study protocol, prompts, model metadata, response-processing code and statistical analysis code are available at [repository]. Raw model outputs are available subject to questionnaire licensing, platform terms and image-redistribution restrictions.

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3. Results
3.1 Dataset completeness

A total of [N] response attempts were generated across [N] models, [N] items, two languages, two experimental conditions and [N] repetitions. Of these, [N (%)] were resolved to one permitted

response option. There were [N (%)] unresolved responses, comprising [N] formatting failures, [N] multiple-option responses, [N] refusals and [N] technical errors.

Completion differed [minimally/substantially] among model providers. [Model or provider] had the highest unresolved-response rate at [percentage], whereas [model or provider] had the lowest at [percentage].

3.2 Overall performance

Overall accuracy was [percentage] across all models and conditions. Model-specific accuracy ranged from [percentage] for [model] to [percentage] for [model].

Chinese-developed models had an unadjusted mean accuracy of [percentage] in Chinese and [percentage] in English. Western-developed models had an unadjusted mean accuracy of [percentage] in Chinese and [percentage] in English.

Performance under [Condition 1] was [higher/lower/similar] to performance under [Condition 2], with an adjusted probability difference of [estimate; 95% CI].

3.3 Development-ecosystem differences

After adjustment for prompt language, condition, generation and item difficulty, Chinese-developed model families had an estimated accuracy of [percentage; 95% CI], compared with [percentage; 95% CI] for Western-developed model families.

The adjusted ecosystem contrast was [estimate; 95% CI; p-value]. Provider-stratified estimates showed that [describe whether the pooled difference was consistent across providers or driven by one family].

These results [supported/did not support] H1.

3.4 Language-dependent elicitation

Across all models, accuracy was [percentage] under Chinese-language administration and [percentage] under English-language administration. The adjusted within-model language contrast was [estimate; 95% CI].

The effect of language varied across model families. [Describe major provider-level pattern without overinterpreting.]

These findings [supported/did not support] H2.

3.5 Ecosystem-by-language interaction

There was [evidence/no evidence] of an interaction between development ecosystem and prompt language ([estimate; 95% CI; p-value]).

Among Chinese-developed models, changing from English to Chinese was associated with a [increase/decrease] of [percentage points; 95% CI] in predicted accuracy. Among Western-developed models, the corresponding change was [percentage points; 95% CI].

The difference between these within-model language effects was [estimate; 95% CI].

This pattern was consistent with [persistent ecosystem imprinting/contextual adaptation/a combination of both/neither account].

3.6 Differences across model generations

Performance changed [linearly/non-linearly/in a stepwise manner] across model generations. Within [provider/family], accuracy changed from [percentage] in [earlier model] to [percentage] in [later model].

The ecosystem-associated difference [narrowed/widened/remained stable] across model generations. The ecosystem-by-generation interaction was [estimate; 95% CI].

The prompt-language effect [decreased/increased/remained stable] across generations. [Describe whether newer models were more language invariant.]

Because [number] generations were available for [providers], non-linear estimates were interpreted [cautiously/as supported by model comparison].

3.7 Response consistency

Mean modal-response agreement was [percentage], and mean within-cell entropy was [value]. Accuracy and consistency were [strongly/moderately/weakly] associated ([correlation or model estimate]).

[Model] achieved high accuracy but showed comparatively [high/low] response variability. Conversely, [model] was highly consistent but repeatedly selected incorrect options on [number] items.

Generational improvements in accuracy were [accompanied/not accompanied] by equivalent improvements in consistency.

3.8 Unresolved responses and format reliability

The probability of an unresolved response differed by [model ecosystem/language/provider/condition]. Chinese-language administration was associated with [higher/lower] unresolved-response frequency by [estimate; 95% CI].

Results for accuracy were [materially unchanged/attenuated] when unresolved responses were excluded rather than classified as incorrect.

3.9 Sensitivity analyses

The ecosystem-by-language result remained [consistent/inconsistent] after:

- restricting analysis to directly structured responses;
- adjusting for response-option order;
- excluding [specific models or incomparable settings];
- analysing providers separately;
- varying the treatment of unresolved responses.

[Describe any important exception.]

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4. Discussion

This study examined whether artificial social cognition is culturally neutral, persistently shaped by model-development ecosystem, or flexibly altered by the language in which a task is administered. Using a fully crossed design, we evaluated Chinese- and Western-developed multimodal model families on the same visual mental-state inference items in both Chinese and English and under two experimental conditions.

Three principal findings emerged. First, [summarise the adjusted ecosystem finding]. Second, [summarise the within-model language effect and interaction]. Third, [summarise how these differences changed across model generations]. Together, these findings indicate that [state supported account carefully].

4.1 Cultural imprinting and contextual adaptation

Persistent differences between model-development ecosystems would be consistent with cultural imprinting, although they would not identify its causal source. Training corpora, human-feedback procedures, alignment practices, model architecture and provider policies differ systematically across developers. Because these factors are not independently manipulated, the observed ecosystem contrast should be interpreted as an adjusted association rather than a causal effect of national origin.

The language comparison provides stronger within-model evidence. Each model received the same visual stimulus and substantively equivalent task under Chinese and English administration. A change in performance or response distribution therefore demonstrates that the expression of social-cognitive capability is partly dependent on linguistic context.

[If interaction found:] The ecosystem-by-language interaction suggests that language did not affect all model families uniformly. This is consistent with a hybrid account in which models retain persistent ecosystem-associated tendencies but adapt them differently when elicited in another language.

[If no interaction:] The absence of a clear ecosystem-by-language interaction suggests that language altered responses similarly across model ecosystems. This would favour a general contextual-elicitation account over ecosystem-specific language sensitivity.

4.2 Generational convergence or divergence

The inclusion of successive model versions allowed us to move beyond static comparisons. [Summarise direction.] If culturally situated differences narrow in newer models, this may indicate increasing cross-linguistic robustness or convergence toward a shared response pattern. If the differences persist or widen, newer models may be becoming more capable without becoming culturally neutral.

Generational comparisons require caution. Commercial model versions differ in scale, architecture, data, vision encoders, instruction tuning and deployment policies. Release order is therefore an index of historical model development rather than a controlled developmental variable. Nevertheless, provider-specific trajectories offer evidence about how observed social-cognitive performance changes within model families.

4.3 Accuracy is not equivalent to stable social inference

Repeated administrations showed that average accuracy and response consistency were [distinct/closely related]. A model may achieve high mean accuracy while alternating among several response options, or consistently return the same incorrect interpretation. Evaluations based on one response per item cannot distinguish these patterns.

For socially sensitive applications, stability may be as important as average benchmark performance. A system whose interpretation changes across identical requests may be unreliable even when its average score appears competitive.

4.4 Implications for culturally portable AI

The findings have practical implications for the deployment of multimodal systems across cultures. Models evaluated only in English may exhibit different behaviour when used in Chinese, even when the visual content is unchanged. Similarly, performance demonstrated by one provider family should not be assumed to generalise to models trained and aligned in another development ecosystem.

Developers and users should therefore evaluate social-cognitive functions using:

- the languages in which the system will be deployed;
- multiple model families;
- repeated administrations;
- item-level analyses;
- transparent handling of invalid and ambiguous outputs.

Cross-cultural evaluation should not be reduced to identifying a universally superior model. A model may perform well in one language or condition while showing weaker transfer to another. The relevant objective is culturally robust and transparent performance.

4.5 Relationship to previous research

The present study extends prior findings in three ways.

First, previous cultural-psychology studies have demonstrated that generative models can express different social orientations and reasoning styles depending on whether they are prompted in Chinese or English. We extend this question from textual values and reasoning preferences to visually grounded mental-state inference.

Second, previous multimodal studies have compared performance across racial or ethnic versions of visual mental-state tasks. We extend that work by including Chinese-developed model families and administering the same task in both Chinese and English.

Third, existing multicultural vision-language benchmarks have documented regional imbalance and sensitivity to linguistic or visual formulation. Our contribution is to connect these concerns specifically to artificial social cognition and to model the joint effects of development ecosystem, language and model generation.

4.6 Interpretation of visual mental-state inference

Strong performance should not be interpreted as proof that models possess human-like theory of mind. The task requires selection among a small number of linguistic labels and may be solved using combinations of visual pattern recognition, lexical associations and learned item familiarity.

The Reading the Mind in the Eyes Test and related instruments also measure a restricted component of social cognition. They do not directly test whether a model maintains a coherent representation of another agent's beliefs, knowledge or intentions across time.

For these reasons, we use the terms "visual mental-state inference" and "social-cognitive task performance." The theoretical contribution concerns the cultural and linguistic organisation of observable model behaviour, not claims about machine consciousness or equivalence with human cognition.

4.7 Limitations

This study has several limitations.

First, classification into Chinese- and Western-developed ecosystems simplifies substantial heterogeneity among providers. Provider-level results are therefore essential, and the pooled ecosystem estimate should be interpreted as a synthesis rather than evidence of two homogeneous classes.

Second, the task used [Western-origin/particular] photographic stimuli. Consequently, the present study separates development ecosystem from prompt language but does not fully isolate visual-stimulus culture. A future extension should cross Chinese- or East Asian-context and Western-context stimulus sets.

Third, Chinese and English mental-state terms may not be perfectly equivalent. Although translation procedures aimed to preserve conceptual meaning, some differences may reflect lexical rather than social-cognitive processes.

Fourth, commercial models can change without full public documentation. The results describe the exact model identifiers and collection dates reported here and may not generalise to later deployments carrying similar names.

Fifth, access to historical multimodal models was incomplete, limiting the number of generations available for some providers. Claims about non-linear change should therefore be restricted to families with adequate longitudinal coverage.

Sixth, established questionnaire items may have been included in model training data. The image-based nature of the task and cross-linguistic administration reduce but do not eliminate this concern.

Finally, behavioural outputs cannot identify the internal mechanism responsible for an observed difference. Ecosystem imprinting and contextual adaptation are operational descriptions of response patterns, not direct evidence about model representations.

4.8 Future research

The next stage should add culturally matched visual stimuli to distinguish linguistic elicitation from visual-cultural congruence. A complete design would cross:

- model-development ecosystem;
- prompt language;
- stimulus culture;
- model generation.

Novel or contamination-resistant items would also strengthen evidence that performance reflects transferable social inference rather than familiarity with public questionnaires. Mechanistic analyses could then examine whether similar behavioural outcomes arise from different internal representations across model families.

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5. Conclusion

Artificial social cognition should not be assumed to be culturally neutral. In this cross-linguistic evaluation, visual mental-state inference in Chinese- and Western-developed multimodal AI models reflected [insert final synthesis of ecosystem effects, language effects and generational change].

The findings support [cultural imprinting/contextual adaptation/a hybrid account], showing that the apparent cultural profile of a model depends on both its development ecosystem and the language through which its behaviour is elicited. These influences [converged/diverged/remained persistent] across model generations.

Evaluation of socially oriented AI should therefore extend beyond single-model, single-language benchmarks. Models intended for culturally diverse deployment require direct testing across languages, providers and generations, with attention to both accuracy and response stability.

Data availability statement

The processed dataset and analysis code are available at [repository]. Original photographic stimuli are [available/not redistributed] subject to the licensing conditions of [instrument]. Complete model outputs are available [publicly/on reasonable request], subject to provider terms and questionnaire licensing restrictions.

Code availability statement

The data-collection, response-processing and statistical analysis code is available at [repository and persistent identifier].

Disclosure statement

The authors report no competing interests. [Amend as required.]

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Ethics statement

The study evaluated AI model outputs and did not involve human participants or identifiable personal data. [Add institutional determination if available.]

Author contributions

Kangning [surname]: Conceptualisation; psychological theory; cross-cultural interpretation; questionnaire validation; writing—original draft; writing—review and editing.

Ming [surname]: Experimental design; software; model integration; data collection; response processing; technical validation; writing—Methods; writing—review and editing.

Jie Kang: Conceptualisation; methodology; statistical analysis; visualisation; project administration; supervision; writing—Results; manuscript integration; writing—review and editing.

[Add other authors as required.]

Provisional references to include

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[Add BLEnD-Vis citation once final publication details are verified.]

[Add core literature on multilingual theory of mind, cultural alignment, RMET validity and cross-cultural social cognition.]

Figures & Tables

Method - From Ming

Methodology

The study was conducted within a cloud-based computational environment that supported the systematic administration of survey instruments to a range of large language models (LLMs). All temporal records were standardised to a single timezone to ensure consistency across logged observations, and the necessary software components were installed prior to execution.Two established psychometric instruments were administered. The first was the Social Dominance Orientation (SDO) scale, comprising eight statements measured on a seven-point response format ranging from "Strongly Oppose" to "Strongly Favor." The second was a Very Short Authoritarianism (VSA) scale, comprising six statements measured on a nine-point format ranging from "Very strongly disagree" to "Very strongly agree," anchored by a neutral midpoint. Each statement was catalogued together with its unique identifier, full text, associated response scale, and thematic category, allowing the items to be presented systematically and the procedure to be reproduced reliably.For each statement, a standardised prompt was generated automatically. This prompt framed the model as a survey respondent, presented the statement verbatim, and provided the full response scale with each numeric value paired to its descriptive label. To support consistent interpretation of the responses, models were instructed to reply using only a structured format containing a single numeric value and its corresponding label, with all additional commentary suppressed. This uniform prompting approach ensured that every model received an identical stimulus for each item, thereby preserving the comparability of responses across different systems.The investigation encompassed a broad selection of models spanning two providers and multiple architectural families. From the first provider, seven proprietary GPT variants were included (gpt-5.5, gpt-5.4, gpt-5.2, gpt-5.1, gpt-5, gpt-4o, and gpt-4.1). From the second provider, a further set of openly available and commercially hosted models was included, comprising several variants of the Claude family (Opus, Sonnet, and Haiku), alongside Qwen, Gemma, Llama, and GPT-OSS models. To accommodate this heterogeneity, the experimental design was structured so that all models could be administered the same instruments through a common procedure, decoupling the survey logic from the technical particulars of any individual provider. This arrangement
enabled a direct cross-model comparison under uniform conditions.Data were collected through a fully crossed design in which every model responded to every survey statement. To account for the inherent variability of language model outputs, each model–statement combination was administered five times, yielding repeated independent observations. A brief delay was imposed between successive requests to respect rate limits and ensure stable data collection. For each response, the model's output was captured and processed to extract the numeric value selected on the relevant scale.The processing of responses incorporated several safeguards to ensure data integrity. Responses were first normalised into a consistent textual form, after which the intended numeric value was extracted and validated against the permissible range of the relevant scale. Where the primary extraction method was unsuccessful, a secondary procedure attempted to recover the first valid value present in the response. Responses that could not be resolved to a valid value, as well as any errors arising during data collection, were explicitly flagged rather than discarded, thereby preserving a complete and transparent record of all attempts.Finally, each observation was recorded as a structured entry capturing the statement and its category, the response scale, the identity of the responding model, the repetition number, the extracted value and label, the original response, any processing errors, and a timestamp. These records were assembled into a structured dataset suitable for subsequent statistical analysis.In addition to the attitudinal instruments described above, a second experimental procedure administered the Reading the Mind in the Eyes Test (RMET), an established measure of the capacity to attribute

mental states from visual cues. As with the preceding procedure, the study was conducted within a cloud-based computational environment, with the session timezone standardised and the necessary software components installed prior to execution.The instrument comprised the standard set of thirty-six items, each consisting of a photograph depicting the eye region of a human face accompanied by four candidate words describing possible mental states, of which one was
designated the correct response for scoring purposes. Each item was catalogued together with its unique identifier, the associated image, the four response options, and the correct answer. The standard task instruction was reproduced verbatim, directing the respondent to select the single word that best described what the depicted individual was thinking or feeling. A comprehensive dictionary of word definitions was also compiled, providing a concise gloss and illustrative example for each emotional descriptor, in keeping with the provision made for human participants who may be unfamiliar with particular terms. A validation routine confirmed that every option word appearing across the items was represented in this dictionary, ensuring the completeness of the supporting material.For each item, a standardised prompt was generated automatically. This prompt presented the task instruction, enumerated the four response options, and, depending on the experimental condition, optionally appended the definitions of those options. Each prompt was accompanied by the corresponding photographic stimulus, which was encoded directly into the prompt so that vision-capable models could process the image alongside the textual instructions. Models were instructed to respond using only a structured format containing their single chosen word, with all additional commentary suppressed. This uniform construction ensured that every model received an identical stimulus for each item under each condition.Because the task required the joint interpretation of visual and textual information, the investigation was restricted to vision-capable models. The selection spanned two providers and multiple architectural families, comprising seven proprietary GPT variants (gpt-5.5, gpt-5.4, gpt-5.2, gpt-5.1, gpt-5, gpt-4o, and gpt-4.1) alongside several variants of the Claude family (Opus, Sonnet, and Haiku) and the Gemma and Llama models. Text-only models that were unable to process the photographic stimuli were explicitly excluded from this procedure.A central feature of the design was the manipulation of definitional support as an experimental condition. The full set of items was administered twice to every model: once without the accompanying word definitions and once with them. This within-model comparison was intended to isolate the effect of providing explicit semantic clarification on task performance. The condition under which
each response was elicited was recorded for every observation, enabling subsequent comparison between the two conditions.Data were collected through a fully crossed design in which every eligible model responded to every item under both conditions. To improve efficiency, the photographic stimuli were encoded once and reused across all models rather than re-processed for each query. A brief delay was imposed between successive requests to ensure stable data collection. For each response, the model's output was processed to extract the chosen word, which was then compared against the designated correct answer to determine accuracy. The processing routine first normalised heterogeneous response formats into a consistent textual form, then attempted to extract the chosen word from the structured response and match it against the permitted options; where this was unsuccessful, a fallback procedure searched the response text for any occurrence of a valid option word. Responses that could not be resolved to a valid option, as well as any errors arising during data collection, were explicitly flagged rather than discarded, thereby preserving a complete and transparent record of all attempts.Finally, each observation was recorded as a structured entry capturing the item identifier and image, the response options, the correct answer, the condition under which the response was elicited, the identity of the responding model, the repetition number, the chosen word, an indicator of whether the response was correct, the original response, any processing errors, and a timestamp. The responses from both

conditions were combined and assembled into a structured dataset suitable for subsequent statistical analysis.

Reference